

Accessibility Scan Analysis and Conclusions – Aegean Homepage

For this assignment, the homepage of Aegean Airlines was tested for accessibility using both manual and automated methods. Manual testing covered keyboard navigation, NVDA screen reader testing, image alternative text review, and a scan with the WAVE extension. Automated testing was handled by Playwright using axe-core. The goal was to identify accessibility problems, compare automated and manual findings, and evaluate the scanner's overall usefulness.

Axe-core Test

I created a Playwright accessibility test targeting the Aegean Airlines homepage (<https://en.aegeanair.com/>), configured to check against the following standards:

- WCAG 2.0 A and AA
- WCAG 2.1 A and AA

The test ran successfully and failed as expected.

Results from the Scanner

The axe-core scan identified two accessibility violations.

1. button-name (*Critical — 2 affected nodes*) Several buttons lacked an accessible name. The issue centred on language selection buttons (Greek and English), which used flag images without any supporting text, aria-label, or aria-labelledby attribute.

2. color-contrast (*Serious — 3 affected nodes*) Three links in the news section had a contrast ratio of 4.16:1, falling short of the WCAG AA minimum of 4.5:1. This makes text harder to read, particularly for users with low vision.

Comparison with Manual Testing

The scanner results overlapped with some manual findings, but the two approaches revealed quite different issues overall.

Where they aligned: The button-name violation corresponded to controls that NVDA also failed to announce clearly, and the contrast issue had already been flagged by WAVE. This confirms that automated tools are reliable for catching technical problems like missing accessible names and insufficient contrast.

Where manual testing went further: Many significant barriers were only discovered through keyboard and screen reader testing. These included unclear visible focus on interactive elements, focus escaping the Login/Register popup, skipped controls inside that popup, payment method icons announced as blank, vague link text such as "Read more" and

"more," missing selected-state feedback in the travel inspiration widget, password eye icons with no meaningful name, passenger controls announced only as "button," and confusing calendar announcements in the booking form.

None of these were detected by axe-core, yet they directly affect how blind and keyboard-only users experience the site.

Conclusion

The axe-core scan found two violations — button-name and color-contrast — both of which were confirmed by manual testing as well. However, the scanner missed a substantial number of barriers that only became apparent through real interaction. Issues with focus management, unclear announcements, unnamed controls, and missing feedback all fall outside what automated tools can reliably detect.

The most effective approach combines both methods: automated scanning for fast, consistent detection of code-level issues, and manual testing to evaluate the actual experience for keyboard and screen reader users. On the Aegean homepage, manual testing provided the more complete picture, while axe-core served as a useful complement for confirming technical violations.

Final Reflection

This assignment reinforced that no single testing method is sufficient on its own. Using keyboard navigation, NVDA, WAVE, and axe-core together produced a stronger and more honest evaluation than any one tool could have achieved alone. Accessibility testing should always include both automated and manual components — the tools find what they can, but only real interaction reveals what users actually encounter.